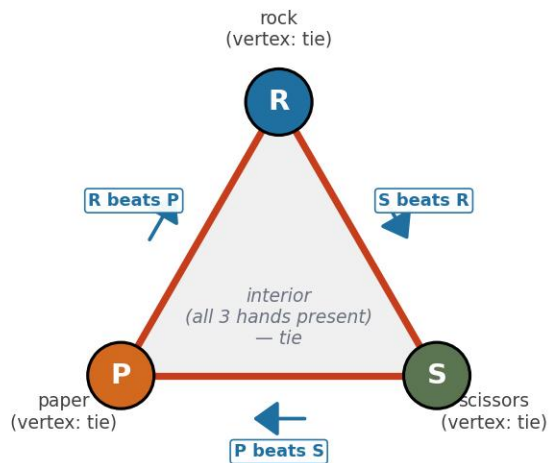
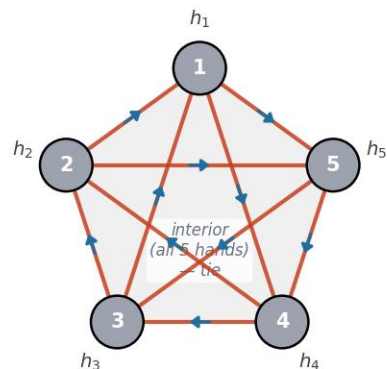


Boundary-driven simplex structure of N-hand cyclic dominance

N = 3: standard rock-paper-scissors
 $k = \lfloor 3/2 \rfloor = 1$ winning gap



N = 5: 5-hand cyclic dominance
 $k = \lfloor 5/2 \rfloor = 2$ winning gaps



$$i \text{ beats } j \iff (j - i) \bmod 5 \in \{1, 2\}$$

vertex (1 hand): tie
 edge (2 hands): decisive 1-face
 interior (all hands): tie

N-hand RPS is a boundary-driven dynamical system: outcomes are decided exclusively on the 1-faces (edges) of the strategy simplex.